UNIT-4: Financial Analysis through Ratios:

Meaning, computation of ratios (i) Liquidity Ratios: Current Ratio and Quick Ratio, (ii) Solvency Ratios: Interest Coverage Ratio and Debt-Equity Ratio, (iii) Activity Ratios: Stock/Inventory Turnover Ratio and Debt Turnover Ratio, (iv) Profitability Ratios: Gross Profit Ratio, Net Profit Ratio & Earning Per Share(EPS) Ratio.

1. Meaning & Purpose of Ratios:

Ratios are among the best known and most widely used tools of financial analysis. Ratio is defined formally as "the indicated quotient of two mathematical expressions". An operational definition of a financial ratio is the relationship between two financial values. The word relationship implies that a financial ratio is the result of comparing mathematically two values.

1.1. Significance of Ratios:

Financial ratios become meaningful to judge financial condition and profitability performance of the company only when there is a comparison. In fact, analysis of ratio involves two types of comparison. First a comparison of present ratio with past and expected future ratios for the same company. When financial ratios for several preceding years are computed, the analyst can determine the composition of change and determine whether there has been an improvement or deterioration in the financial position of the company. It involves comparing the ratios of the company with those similar type of company or with industry averages at the same point of time. Such a comparison would provide considerable insight into the relative financial condition and performance of the company. Hence, financial ratios are highly significant to fulfil the following requirements:

A. Analysis of financial statements
B. Helps in understanding the profitability of the company
C. Analysis of operational efficiency of the firms
D. Checking and maintaining Liquidity of the firms
E. Helps in identifying the financial risks of the company
F. For planning and future forecasting of the firm
G. To compare the performance of the firms

1.2. Users of Financial Ratios:

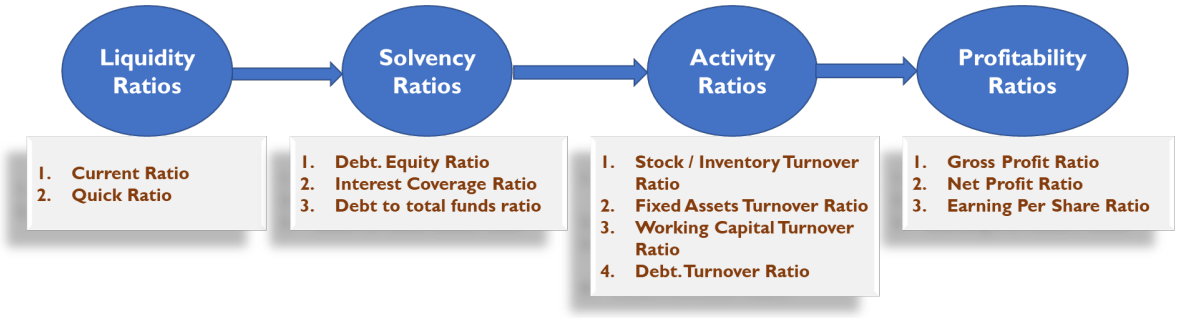
The internal and external stakeholders of a firm such as, employees, shareholders, investors, suppliers, audit and tax authorities, Government, etc. are the people who are interested to know the up to date financial position to fulfil their own requirements. The periodical assessment of ratios are always transparent to all the above stakeholders. These are essential

outcomes to take decisions by each stakeholder in the process of maintaining relationship with the company.

2. Computation of ratios:

Ratios as tools of measuring liquidity, profitability, efficient and financial position of a company can be classified into different categories as follows:

2.1. Classification of Ratios:



2.1.1: Liquidity Ratios:

Liquidity ratios provide test to measure the ability of the corporation to cover its short-term obligations out of its short-term resources. Interpretation of liquidity ratios provide considerable insight into the present cash solvency of the corporation and its ability to remain solvent in times of adversities. Two commonly used liquidity ratios are current ratio and quick ratio.

A. Current Ratio: Current ratio expresses relationship between current assets and current liabilities. It is computed by dividing current assets by current liabilities. The ratio is calculated on the basis of the following formula:

$$Current\ Ratio = \frac{Current\ assets}{Current\ liabilities}$$

Current Assets (Those assets which are converted into cash within a year)	**Current Liabilities** (Those liabilities which are to be paid within a year)
• Cash in hand • Cash at bank • Raw materials • Semi-finished goods • Finished goods • Debtors	• Creditors • Bills payable • Outstanding expenses • Short-term loans • Bank overdraft

• Bills receivables • Prepaid expenses • Short-term investments	

A current ratio of 2:1 is considered as minimum in a sound business. A higher current ratio explains that the company will be able to pay its debts maturing with in a year. On the other hand, a low current ratio points to the possibility that the company may not be able to pay its short-term debts. However, the management point of view higher current ratio is indicative of poor planning since an excessive amount of funds be idle. On the contrary a low ratio would mean inadequacy of working capital which may dealt smooth functioning of the enterprise. Employees, suppliers, creditors are especially interested in knowing the liquidity position of enterprise.

B. Liquid Ratio / Quick Ratio / Quick Ratio: It is a measure of judging the immediate ability of the company to pay off its current obligations. It is obtaining by dividing quick current assets by current liabilities. Quick current assets would comprise those assets which can be liquidated immediately and at minimum loss in order to meet pressing financial obligations. Quick assets consist of:

1. Cash,
2. Marketable securities and
3. accounts receivables.

These are called liquid assets, because they can be converted into cash very shortly.

Quick / Liquid assets = Current assets – [stock + Prepaid expenses]

The following formula is used to calculate the ratio:

$$Quick/Liquid\ Ratio = \frac{Liquid\ assets}{Current\ liabilities}$$

or

$$Current\ Ratio = \frac{Current\ assets - [Stock + Prepaid\ expenses]}{Current\ liabilities}$$

A quick ratio of 1:1 is usually considered adequate.

2.1.2: Solvency Ratios:

These ratios are generally designed to measure the contribution of the company's owners in relations to the funds provided by its credits. These ratios are computed to solicit information along the following lines: (1) the company's ability to cover all it's obligations including short term and long term obligations.(2) the margin of safety afforded to the creditors.(3) the extent

of control of the stock holders over enterprise, and (4) the potential earnings from the use of loan funds. The solvency or leverage ratios are:

A. Debt-Equity Ratio: this is a measure of owner's stake in the business. The proprietors may desire more of funds to be from borrowings because it's reduces owner's risk and interest on borrowed funds is allowed as an expenditure in computing taxable profits but not dividend on shares. The tax is computed on the profits before any dividend is declared. Generally, a ratio of 2:1 is considered normal but in certain cases relaxations are allowed.

$$Debt - Equity\ Ratio = \frac{long\ term\ liabities}{Equity}\ or\ \frac{DEBT}{EQUITY}$$

Long term liability = long term loans

Equity = Equity share capital + reserves + irredeemable preference share capital.

B. Interest-Coverage Ratio: this ratio shows how many times the profit covers the interest. It shows the margin of cover to lender of the company. It always to have profit more than the interest payable. In-case profit is either equal or lesser than interest, the position will be unsafe and it will show that there is nothing left for the share-holders and the position of lenders is also unsafe.

So, this ratio ascertains whether the company is capable of meeting interest on loans easily out of profits or not. The net income of the company should ideally be 6 or 7 times of the fixed interest charges. This ratio is calculated on the basis of the following formula:

$$Debt\ Service/Interest\ Ratio = \frac{net\ profit\ before\ interest\ and\ taxes}{fixed\ interest\ charges}$$

C. Debt to total funds/Total debt total assets Ratio: this ratio compares the owner's contribution with the total outside liabilities of the concern.

$$Debt\ to\ total\ funds\ \ Ratio = \frac{term\ liabities + current\ liabilities}{Equity}$$

This ratio should be watched for 3 to 5 years to see its trend, if declining or increasing. A declining trend in the ratio is a welcome sign as it shows that the company is augmenting its own sources by ploughing back profits or by reducing its dependence on outside borrowing by repaying them. On the other hand an increasing trend in the ratio should be carefully looked into by the banker. A ratio of 4:1 is considered normal.

2.1.3: Activity Ratios:

Activity ratios reflect how efficiently the company is managing its resources. These ratios express relationship between the level of sales and the investment in various assets, viz, inventories, receivables, fixed assets, etc. the important turnover ratios are:

A. Stock or Inventory Turnover Ratio: This ratio measures how many times on an average stock is sold during the year. It also shows efficiency of the firm. Lower inventory turnover ratio shows that the stock is blocked and not immediately sold. It shows the poor performance of the business and inefficiency of the management. The ratio measures the effectiveness of the stock policy of the management. It should be constant effort of the management to dispose o stock as the earliest.

$$Stock\ Turnover\ Ratio = \frac{Cost\ of\ goods\ sold}{Average\ Stock}$$

Cost of goods sold = [Opening stock + Purchases + Direct expenses] – Closing Stock

(or)

Sales – Gross Profit

Average Stock = $\frac{Opening\ Stock + Closing\ Stock}{2}$

B. Fixed Assets Turnover Ratio: Fixed assets are used in the business for producing goods to be sold. The effective utilization of fixed assets will result in increased production and reduced cost. It also ensures whether investment in the assets have been judicious or not. The following formula is used for measurement of the ratio:

$$Fixed\ Assets\ Turnover\ Ratio = \frac{Sales\ or\ cost\ of\ sales}{Fixed\ Assets}$$

C. Working Capital Turnover Ratio: This ratio measures the relationship between working capital and sales. The ratio shows the number of times the working capital results, in a sales. Working capital as usual is the excess of current assets over the current liabilities. The following formula is used measures this ratio:

$$Working\ capital\ turnover\ ratio = \frac{Sales\ or\ cost\ of\ sales}{Working\ Capital}$$

D. Debtors Turnover Ratio: It is also known as receivables turnover ratio. It establishes relationship between credit sales and average debtor. This ratio is calculated on the basis of the following formula:

$$Debtos\ turnover\ ratio = \frac{Credit\ sales}{Average\ Debtors}$$

Credit Sales = It is given separately, or it may be the difference between total sales and cash sales.

Average Debtors = $\frac{Opening\ Debtors+Closing\ Debtors}{2}$

2.1.4: Profitability Ratios:

Profitability ratios are as a matter of fact, best indicators of overall efficiency of the business, concern because they compare return or value over and above the values put into a business with sale or service carried on by the enterprise with the help of assets employed. Thus profitability ratios are of two types. Profitability as related to sales and profitability as related to investment.

A. Gross Profit Ratio: This ratio establishes relationship between gross profit and sales to measure the relative operating efficiency of the corporation and to reflect its pricing policies. It is computed by dividing sales into sales minus cost of goods sold. Sometimes, it is calculated by taking cost of goods sold instead of sales. It indicates the position of trading result.

$$Gross\ Profit\ ratio = \frac{Gross\ Profit}{Net\ Sales}$$

B. Net Profit Ratio: It is also called net profit margin ratio. It is calculated by dividing net income (after tax) by net sales. This ratio provides considerable insight into the overall efficiency of the business. A higher ratio is an indication of efficient use of total resources. A low ratio on the contrary would mean poor financial planning and low efficiency. The ratio can be calculated with the help of the following formula:

$$Net\ Profit\ ratio = \frac{Net\ Profit}{Net\ Sales}$$

Profitability as related to investment: this category of profitability ratio includes return on total capital employed and return on net worth.

(a) Return on investment or capital employed: This is one of the most important profitability ratios. It indicates the relation on net profit with capital employed in the business. Net profit for calculating return on investment will mean the net profit before interest, tax and dividend. Capital employed means long-term funds which consists of the following:

Capital employed = [Equity share capital + Preference share capital + Reserves + accumulated profit + Long term loans] – Fictitious Assets like underwriting commission, Preliminary expenses and non-operating assets like investment.

(or)

= Net Fixed assets + Working capital

Net Fixed assets = Fixed assets – Depreciation

Working capital = Current Assets – Current liabilities and provisions.

Return on Investment can be calculated on the basis of the following formula(ROI):

$$= \frac{Net\ Profit\ before\ tax, Interest\ and\ Dividend}{Capital\ Employed}$$

C. Earning Per-share Ratio: This ratio is computed by dividing earnings available to the equity share holders by the total number of equity shares outstanding.

$$EPS = \frac{Earning\ after\ tax - Preferred\ dividents(if\ any)}{Equity\ Shares\ Outstanding}$$

This ratio reveals the amount of the periods earnings after taxes which access to each equity share. This ratio is an important index because it indicates whether the wealth of each shareholder on a per-share basis has changed over the period. A year – to – year comparison of earnings per share can be very informative to investors. Thus, EPS provides an insight into the investment potentiality of an organization.

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Problem 1:

The following is the Balance Sheet of a company as on 31st March:

Liabilities	*Rs.*	*Assets*	*Rs.*
Share Capital	2,00,000	Land and Buildings	1,40,000
Profit & Loss Account	30,000	Plant and Machinery	3,50,000
General Reserve	40,000	Stock	2,00,000
12% Debentures	4,20,000	Sundry Debtors	1,00,000
Sundry Creditors	1,00,000	Bills Receivable	10,000
Bills Payable	50,000	Cash at Bank	40,000
	8,40,000		8,40,000

Calculate :

(1) Current Ratio
(2) Quick Ratio
(3) Inventory to working Capital
(4) Debt to Equity Ratio
(5) Proprietary Ratio
(6) Capital Gearing Ratio
(7) Current Assets to Fixed Assets

SOLUTION :

(1) Current Ratio $= \dfrac{\text{Current assets}}{\text{Current Liabilities}}$

$= \dfrac{\text{Rs. } 3,50,000}{\text{Rs. } 1,50,000} = \mathbf{2.33 : 1}$

(2) Quick Ratio $= \dfrac{\text{Liquid Assets}}{\text{Liquid Liabilities}}$

$= \dfrac{\text{Rs. } 1,50,000}{\text{Rs. } 1,50,000} = \mathbf{1 : 1}$

(3) Inventory to Working Capital $= \dfrac{\text{Inventory}}{\text{Working Capital}}$

$= \dfrac{\text{Rs. } 2,00,000}{\text{Rs. } 2,00,000} = \mathbf{1 : 1}$

(Working Capital = Current Assets – Current Liabilities
= Rs. 3,50,000 – Rs. 1,50,000 = Rs. 2,00,000)

(4) Debt to Equity Ratio $= \dfrac{\text{Long Term Debts}}{\text{Shareholders' Fund}}$

$$= \frac{\text{Rs. } 4,20,000}{\text{Rs. } 2,70,000} = \mathbf{1.56 : 1}$$

(Or)

$$= \frac{\text{Long Term Debts}}{\text{Shareholders' Fund + Long Term Debts}}$$

$$= \frac{\text{Rs. } 4,20,000}{\text{Rs. } 2,70,000 + 4,20,000} = \mathbf{0.6 : 1}$$

(5) Proprietary Ratio $= \frac{\text{Shareholders' Fund}}{\text{Total Assets}}$

$$= \frac{\text{Rs. } 2,70,000}{\text{Rs. } 8,40,000} = \mathbf{0.32 : 1}$$

(6) Capital Gearing Ratio $= \frac{\text{Fixed Interest Bearing Securities}}{\text{Equity Share Capital}}$

$$= \frac{\text{Rs. } 4,20,000}{\text{Rs. } 2,00,000} = \mathbf{2.1 : 1}$$

(7) Current Assets to Fixed Assets Ratio $= \frac{\text{Current Assets}}{\text{Fixed Assets}}$

$$= \frac{\text{Rs. } 3,50,000}{\text{Rs. } 4,90,000} = \mathbf{0.71 : 1}$$

Problem 2:

From the following particulars found in the Trading, Profit and Loss Account of A Company Ltd., work out the operation ratio of the business concern:

TRADING ACCOUNT OF A COMPANY LTD.
for the period ending December 31

Dr. *Cr.*

Expenses	*Rs.*	*Incomes*	*Rs.*
To Opening Stock	1,400	By Net Sales	10,000
To Purchases	6,400	By Closing Stock	600
To Direct Expenses	300		
To Gross Profit	2,500		
	10,600		10,600

PROFIT AND LOSS ACCOUNT OF A COMPANY LTD.
for the period ending December 31

Dr. *Cr.*

Expenses	*Rs.*	*Incomes*	*Rs.*
To Operating Expenses:		By Gross Profit	2,500
(a) Administrative Expenses	1,600		
(b) Selling and Distribution Expenses	300		
To Financial Expenses	100		
To Net Profit	500		
	2,500		2,500

SOLUTION :

$$\text{Operating Ratio} = \frac{\text{Cost of goods sold and other operating expenses}}{\text{Net sales}} \times 100$$

	Rs.
Cost of Goods Sold :	
Opening Stock	1,400
Purchases	6,400
Direct Expenses	300
	8,100
Less Closing Stock	600
Cost of Goods Sold	7,500
Operating Expenses :	Rs.
(a) Administrative Expenses	1,600
(b) Selling and Distribution Expenses	300
(c) Financial Expenses	100
Operating Expenses	2,000

$$\text{Operating Ratio} = \frac{7,500 + 2,000}{10,000} \times 100 = 95\%$$

Problem 3:

The following is the summarised Profit and Loss Account of Taj Products Ltd. for the year ended 31st December:

PROFIT AND LOSS ACCOUNT

	Rs.		Rs.
Opening Stock of Materials	99,500	Sales	8,50,000
Purchase of Materials	3,20,000	Stock of Materials (Closing)	89,000
Direct Wages	2,25,250		
Manufacturing Expenses	14,250	Stock of Finished Goods (Closing)	60,000
Selling & Distribution		Non-operating Income Interest	3,000
Expenses	30,000	Profit on Sale of Shares	6,000
Administrative Expenses	1,50,000		
Finance Charges	15,000		
Non-operating Expenses:			
Loss on Sale of Assets	4,000		
Net Profit	1,50,000		
	10,08,000		10,08,000

Work out the following ratios :

(1) Gross Profit Ratio
(2) Net Profit Ratio
(3) Operating Ratio
(4) Cost Ratios (to cost of Production)
  (*i*) Materials Consumed Ratio
  (*ii*) Labour Cost Ratio
  (*iii*) Production Overhead Cost Ratio.

SOLUTION:

		Rs.
Gross Sales (*a*)		8,50,000
Less : Cost of Goods Sold :	Rs.	
Opening Stock of Materials	99,500	
Add : Materials Purchased	3,20,000	
	4,19,500	
Less : Stock of Materials (Closing)	89,000	
Materials Consumed : (*b*)	3,30,500	
Direct Wages	2,25,250	
Manufacturing Expenses	14,250	
Cost of Production (*c*)	5,70,000	
Less : Closing Stock of Finished Products	60,000	
Cost of Goods Sold (*d*)		5,10,000
Gross Profit (*e*)		3,40,000
Less : Administrative Expenses	1,50,000	
Selling and Distribution	30,000	1,80,000
Net Operating Profit before Interest and Taxation : (*f*)		1,60,000
Add : *Non-operating Incomes* (*g*)		
Interest	3,000	
Profit on Sale of Shares	6,000	9,000
		1,69,000
Less : Loss on Sale of Assets	4,000	
Finance Charges	15,000	19,000
Income before Taxation (*h*)		1,50,000

(1) Gross Profit Ratio $= \frac{\text{Gross Profit}}{\text{Net Sales}} \times 100$

$= \frac{\text{Rs. } 3,40,000}{\text{Rs. } 8,50,000} \times 100 = 40\%$

(2) Net Profit Ratio $= \frac{\text{Net Profit}}{\text{Net Sales}} \times 100$

$= \frac{1,60,000}{\text{Rs. } 8,50,000} \times 100 = 18.82\%$

(3) Operating Ratio $= \frac{\text{Cost of Goods Sold + Operating Expenses}}{\text{Net Sales}} \times 100$

$= \frac{\text{Rs. } 5,10,000 + \text{Rs. } 1,80,000}{\text{Rs. } 8,50,000} \times 100 = 81.18\%$

(4) (*i*) Material Consumed Ratio $= \frac{\text{Material Consumed}}{\text{Cost of Product}} \times 100$

$= \frac{\text{Rs. } 3,30,500}{\text{Rs. } 5,70,000} \times 100 = 57.98\%$

$$(iii)\ \text{Production Overhead Ratio} = \frac{\text{Production Overhead}}{\text{Cost of Production}} \times 100$$

$$= \frac{\text{Rs. } 14{,}250}{\text{Rs. } 5{,}70{,}000} \times 100 = 2.5\%$$

Problem 4:

From the following Balance Sheet and additional information, you are required to calculate:

(i) Return on Total Resources

(ii) Return on Capital Employed

(iii) Return on Shareholders' Fund

BALANCE SHEET as on 31st Dec.

	Rs.		*Rs.*
Share Capital (Rs. 10)	8,00,000	Fixed Assets	10,00,000
Reserves	2,00,000	Current Assets	3,60,000
8% Debentures	2,00,000		
Creditors	1,60,000		
	13,60,000		13,60,000

Net operating profit before tax is Rs. 2,80,000. Assume tax rate at 50%. Dividend declared amounts to Rs.1,20,000. *(B.Com. MS.)*

SOLUTION:

$$(i)\ \text{Retrun on Total Resources} = \frac{\text{Profit after Tax}}{\text{Total Assets}} \times 100$$

$$= \frac{\text{Rs.1,40,000}}{\text{Rs.13,60,000}} \times 100 = 10.29\%$$

(*ii*) Return on Capital Employed

$$= \frac{\text{Profit before Tax \& Interest}}{\text{Capital Employed}} \times 100$$

$$= \frac{\text{Rs.2,96,000}}{\text{Rs.12,00,000}} \times 100 = 24.7\%$$

(*iii*) Return on Shareholders' Fund

$$= \frac{\text{Profit after Tax}}{\text{Shareholders Fund}} \times 100$$

$$= \frac{\text{Rs.1,40,000}}{\text{Rs.10,00,000}} \times 100 = 14\%$$

Problem 5:

A company has capital of Rs. 10, 00,000; its turnover is 3 times the capital and the margin on sales is 6%. What is the return on investment.

SOLUTION :

$$\text{Capital Turnover Ratio} = \frac{\text{Sales}}{\text{Capital}}$$

$$3 = \frac{\text{Sales}}{\text{Rs. 10,00,000}}$$

$$\text{Sales} = \text{Rs. 30,00,000}$$

$$\text{Rate of Return on Investment} = \frac{\text{Gross Profit}}{\text{Investment}} \times 100$$

$$= \frac{\text{Rs. 1,80,000}}{\text{Rs. 10,00,000}} \times 100$$

$$= 18\%$$

$$\text{Gross Profit} = 6\% \text{ of Rs. } 30,00,000$$

$$= \text{Rs. } 1,80,000$$

Problem 6:

Ram & Company supplies you the following information regarding the year ended 31st December:

Cash Sales	Rs. 80,000
Credit Sales	Rs. 2,00,000
Return Inward	Rs. 10,000
Opening Stock	Rs. 25,000
Closing Stock	Rs. 30,000

Gross Profit Ratio is 25%

Find out Inventory Turnover. *(B.Com. Madurai)*

SOLUTION :

$$\text{Inventory Turnover} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory at Cost}}$$

$$\text{Net Sales} = \text{Rs. } 80,000 + \text{Rs. } 2,00,000 - \text{Rs. } 10,000$$

$$\text{Rs.} = 270,000$$

$$\text{Cost of Goods Sold} = \text{Sales} - \text{Gross Profit}$$

$$= \text{Rs. } 2,70,000 - (25\% \text{ of Rs. } 2,70,000)$$

$$= \text{Rs. } 2,70,000 - \text{Rs. } 67,500$$

$$= \text{Rs. } 2,02,500$$

$$\text{Average Inventory} = \frac{\text{Rs. } 25,000 + \text{Rs. } 30,000}{2}$$

$$= \text{Rs. } 27,500$$

$$\text{Inventory Turnover} = \frac{\text{Rs. } 2,02,500}{\text{Rs. } 27,500} = 7.36 \text{ Times}$$

Significance:

A high Inventory turnover ratio is better than a low ratio. A high ratio implies good inventory 'management and an indication of under-investment. It will adversely affect the ability of a firm to meet customers' demand. At the same time, a higher ratio reflects efficient business activities.

A low inventory turnover ratio is dangerous. It is an indication of excessive inventory and over investment in inventory. A low ratio may be result of inferior quality goods, stock of un-saleable and absolute goods. A lower ratio reflects dull business and suggests that some steps should be taken to push up sales.

Problem 7:

The following is the Profit and Loss Account of Burn Ltd.

	Rs.		Rs.
To Opening Stock :			
Materials	60,000	By Sales	12,70,000
Finished Goods	80,000	By Closing Stock :	
		Materials	70,000
To Purchase (Raw Materials)	6,00,000	Finished Goods	1,00,000
To Wages	3,00,000		
To Factory Expenses	1,00,000		
To Other Expenses	2,00,000		
To Net Profit	1,00,000		
	14,40,000		14,40,000

You are required to calculate the following :

(*i*) Stock Turnover

(*ii*) Raw Materials Turnover

(*iii*) Average Material Holding *(B.Com. Bhopal, Agra, Madurai)*

SOLUTION :

$$(i)\ \text{Stock Turnover} = \frac{\text{Cost of Sales}}{\text{Average Stock}} = \frac{\text{Rs. } 11,70,000}{\text{Rs. } 90,000} = \mathbf{13\ times}$$

$$(ii)\ \text{Raw Materials Turnover} = \frac{\text{Raw Material Consumed}}{\text{Average Stock of Raw Materials}}$$

$$= \frac{\text{Rs. } 5,90,000}{\text{Rs. } 65,000} = \mathbf{9.08\ times}$$

$$(iii)\ \text{Average Material Holding} = \frac{365}{\text{Raw Material Turnover}}$$

$$\text{(Turnover Period)} = \frac{365}{9.08} = \mathbf{40\ days}$$

Workings :

(1) Cost of Sales

= Total Cost + Opening Stock of Finished Goods – Closing Stock of Finished Goods

= Rs. 11,90,000 + Rs. 80,000 – Rs. 1,00,000 = Rs. 11,70,000

(2) Average Stock $= \frac{\text{Opening Stock + Closing Stock}}{2}$

Average Stock of Finished Goods

$= \frac{\text{Rs. 80,000 + Rs. 1,00,000}}{2} = \text{Rs. 90,000}$

Average Stock of Raw Materials

$= \frac{\text{Rs. 60,000 + Rs. 70,000}}{2} = \text{Rs.65,000}$

(3) Raw Materials Consumed

= Opening Stock + Purchase – Closing Stock

= Rs. 60,000 + Rs. 6, 00,000

= Rs. 5, 90,000

Problem 8:

Calculate Debtors Velocity from the following details:

Opening Balance of Debtors Rs. 10,000

Credit Sales during the year Rs. 20,000

Sales Returns Rs. 1,000

Discount on Sales Rs. 50

Cash collected from Debtors during the year Rs, 5,000

Bad Debts Rs. 500

Bad Debt Provision at 10%

SOLUTION :

The following components are needed to find Debtors Velocity :

1. Net Sales = Gross Sales – (Sales Returns + Discount on Sales)
   = Rs. 20,000 – (Rs. 1,000 + Rs. 50)
   = Rs. 18.950
2. Net Debtors on Closing = Opening Debtors
   Add : Credit Sales
   Less : Cash Collection
   Less : Sales Returns
   Less : Sales Discount

Rs. 10,000 + Rs. 20,000 – (Rs. 5,000 + Rs. 1,000 + Rs. 50) = Rs. 23,950

3. Average Debtors $= \frac{\text{Opening Debtors + Closing Debtors}}{2}$

$= \frac{\text{Rs. 10,000 + Rs. 23,950}}{2} = \text{Rs. 16,975}$

4. Average Daily Sales (credit) $= \frac{\text{Rs.18,950}}{365} = \text{Rs.52}$

1. Debtors Turnover (Time)

$$= \frac{\text{Net Sales}}{\text{Average Debtors}} = \frac{\text{Rs. 18,950}}{\text{Rs.16,975}} = 1.12 \text{ Times}$$

2. Collection Period $= \frac{365}{1.12} = 326$ days

3. Debtors Turnover (Percentage)

$$= \frac{\text{Average Dettors}}{\text{Net Sales}} = \frac{\text{Rs. 16,975}}{\text{Rs. 18,950}} \times 100 = 89.58\%$$

4. Collection Period (Days)

= Debtors Turnover (percentage) × 365 = 89.58 × 365 = 326 Days

(or) 5. Collection Period

$$= \frac{\text{Average Debtors}}{\text{Average Daily Sales}} = \frac{\text{Rs. 16.975}}{\text{Rs. 52}} = 326 \text{ Days}$$

Note:

Bills Receivable from the buyer of fixed assets, should be excluded.

Bad and doubtful debts and their provisions are not deducted from the total debtors in order to avoid the impression that a larger amount of receivables have been collected.

Significance:

A turnover ratio of 8 signifies that debtors get converted into cash 8 times in a year. The average collection period of 1.5 months implies that debtors are collected in 45 days. The average collection period ratio measures the quality of debtors since it indicates the rapidity or slowness of their collectability. The shorter the average collection period, the better the quality of debtors.

The higher the Turnover Ratio and the shorter the average collection period, the better the trade credit management and the better the liquidity of debtors. That is, high Turnover Ratio and short collection period imply prompt payment on the part of debtors. On the other hand, low Turnover Ratio and long collection period reflects that payments by debtors are delayed.

Problem 9:

The Capital of a Company is as follows:

	Rs.
9% Preference Shares of Rs.10 each	3,00,000
Equity Shares of Rs.10 each	8,00,000
	11,00,000

The Accountant has ascertained the following information :

Profit (after tax at 60%) Rs. 2,70,000.

Depreciation Rs. 60,000

Equity Dividend Paid 20%

Market Price of equity share Rs. 40.

You are required to state the following, showing the necessary workings :

(*a*) Dividend yield on the equity shares.

(*b*) Cover for the preference and equity dividends.

(*c*) Earnings for equity shares.

(*d*) Price-earnings ratio (*CA*)

SOLUTION :

(*a*) *Dividend yield on the Equity Shares :*

$$= \frac{\text{Dividend per Share}}{\text{Market Price per Share}} \times 100$$

$$= \frac{\text{Rs. 2 (}i.e.\text{ 20\% of Rs. 10)}}{\text{Rs. 40}} \times 100 = \mathbf{5\%}$$

(*b*) *Dividend Coverage Ratio :*

(*i*) *Preference Shares* $= \frac{\text{Profit After Taxes}}{\text{Dividend Payable to Pref. Shareholders}}$

$$= \frac{\text{Rs. 2,70,000}}{\text{Rs. 27,000}} = \mathbf{10\ Times}$$

(Rs. 27,000 is 9% of Rs. 3,00,000)

(*ii*) *Equity* $= \frac{\text{Profit After Taxes – Pref. Share Dividends}}{\text{Dividend Payable to Equity Shareholders at Rs. 2 per Share}}$

$$= \frac{\mathbf{Rs.\ 2,70,000 - Rs.27,000}}{\mathbf{Rs.\ 1,60,000\ OR\ (80,000 \times Rs.\ 2)}}$$

$$= \frac{\text{Rs. 2,43,000}}{\text{Rs. 1,60,000}} = \mathbf{1.518\ Times}$$

(*c*) *Earnings for Equity Shares :*

$$= \frac{\text{Earnings available to Equity Shareholders}}{\text{Number of Equity Shares}}$$

$$= \frac{\text{Rs. 2,43,000}}{\text{80,000}} = \mathbf{Rs.\ 3.0375\ per\ share}$$

(*d*) *Price-earning Ratio :*

$$= \frac{\text{Market Price Per Share}}{\text{Earnings per Share}}$$

$$= \frac{\text{Rs. 40}}{\text{Rs. 3.0375}}$$

$$= \mathbf{13.168\ Times}$$

Problem 10:

Assume that a firm has owners' equity of Rs. 1, 00,000. The ratios for the firm are:

Current Debt to Total Debt	0.40
Total Debt to owners' equity	0.60
Fixed Assets to owners' equity	0.60
Total Assets Turnover	2 Times
Inventory Turnover	8 Times

Complete the following Balance Sheet, given the information above.

Equities	*Rs.*	*Assets*	*Rs.*
Current Debt	.....	Cash	.....
Long-Term Debt	.....	Inventory	.....
Total Debt	.....	Total Current Assets	.....
Owners' Equity	.....	Fixed Assets	.....
Total Equity	.....	Total Assets	.....

SOLUTION.

Equities	*Rs.*	*Assets*	*Rs.*
Current Debt	24,000	Cash	60,000
Long-Term Debt	36,000	Inventory	40,000
Total Debt	60,000	Total Current Assets	1,00,000
Owners' Equity	1,00,000	Fixed Assets	60,000
Total Equity	1,60,000	Total Assets	1,60,000

Workings :

1. Total Debt = 0.60 × Owners' Equity
   = 0.60 × Rs.1,00,000
   = Rs. 60,000
2. Fixed Assets = 0.60 × Owners' Equity
   = 0.60 × Rs.1,00,000
   = Rs. 60,000
3. Total Equity = Total Debt + Owners' Equity
   Rs. 60,000 + Rs. 1,00,000
   = Rs. 1,60,000
4. Total Assets = Current Assets + Fixed Assets (Must be equal to
   = Rs. 1,60,000). Fixed Assets = Rs. 60,000
   Current Assets = Rs. 160,000 – Rs. 60,000 = Rs. 1,00,000
5. Sales :

$$\text{Asset Turnover or } 2 = \frac{\text{Sales}}{\text{Assets}}$$

$$2 = \frac{\text{Sales}}{\text{Rs. 1,60,000}}$$

Sales = Rs. 160,000 × 2 = Rs. 3,20,000

6. Inventory :

$$\text{Inventory Turnover or } 8 = \frac{\text{Sales}}{\text{Inventory}}$$

$$8 = \frac{\text{Rs. } 3,20,000}{\text{Inventory}}$$

$$\text{Inventory} = \frac{\text{Rs. } 3,20,000}{8} = \text{Rs. } 40,000$$

7. Cash = Current Assets – Inventory
= Rs. 1,00,000 – Rs. 40,000 = Rs. 60,000

8. Current Debt = 0.40 × Total Debt = 0.40 × Rs. 60,000 = Rs. 24,000

9. Long-Term Debt = Total Debt – Current Debt = Rs. 60,000 – Rs. 24,000 = Rs. 36,000

Problem 11:

With the following ratios and further information given below, prepare a Trading, Profit and Loss Account and Balance Sheet:

Gross Profit Ratio	25%
Net Profit Ratio	20%
Stock Turnover Ratio	10
Net Profit/Capital	1/5
Capital to Total Liabilities	1/2
Fixed Assets/Capital	5/4
Fixed Assets/Total Current Assets	5/7
Fixed Assets	Rs.10,00,000
Closing Stock	Rs. 1,00,000

SOLUTION :

TRADING AND PROFIT AND LOSS ACCOUNT

	Rs.		*Rs.*
To Opening Stock	20,000	By Sales	8,00,000
To Purchases (Balancing Figure)	6,80,000	By Closing Stock	1,00,000
To Gross Profit	2,00,000		
	9,00,000		9,00,000
To Expenses (Balancing Figure)	40,000	By Gross Profit	2,00,000
To Net Profit	1,60,000		
	2,00,000		2,00,000

BALANCE SHEET As on.....

Liabilities	*Rs.*	*Assets*	*Rs.*
Capital	6,40,000	Fixed Assets	10,00,000
Net Profit	1,60,000	Closing Stock	1,00,000
Liabilities	16,00,000	Current Assets	13,00,000
	24,00,000		24,00,000

Workings :

1. *Capital* :

$$\frac{\text{Fixed Assets}}{\text{Capital}} = \frac{5}{4}$$

$$\text{Capital} = \text{Rs. } 10,00,000 \times \frac{4}{5} = \textbf{Rs. 8,00,000}$$

2. *Liabilities* :

$$\text{Capital} = \frac{1}{2} \text{ of Total Liabilities}$$

$$\text{Liabilities} = \text{Rs. } 8,00,000 \times \frac{2}{1} = \textbf{Rs. 16,00,000}$$

3. Net Profit $= \frac{1}{5}$ of Capital $= \text{Rs. } 8,00,000 \times \frac{1}{5} = \textbf{Rs. 1,60,000}$

4. Net Profit = 20% of sales = Rs. 1,60,000

So, Sales = $\text{Rs. } 1{,}60{,}000 \times \frac{100}{20}$ = **Rs. 8,00,000**

5. Gross Profit = 25% of Sales = 25% of Rs. 8,00,000 = **Rs. 2,00,000**

6. *Opening Stock* :

Stock Turnover = 10

Cost of Sales = Rs. 8,00,000 – Rs. 2,00,000 = Rs. 6,00,000

Average Inventory = Rs. 60,000

Opening Stock = Rs. 60,000 × 2 = Rs. 1,20,000 – Rs. 1,00,000 = **Rs. 20,000**

7. *Current Assets* :

Fixed Assets/Total Current Assets = 5/7

Current Assets = $\text{Rs. } 10{,}00{,}000 \times \frac{7}{5}$ = Rs. 14,00,000

Other Current Assets = Current Assets – Stock

= Rs. 14,00,000 – Rs. 1,00,000

= **Rs. 13,00,000**

Problem 12:

Extract from financial accounts of X, Y, Z Ltd. are:

	Year I		*Year II*	
	Assets Rs.	*Liabilities Rs.*	*Assets Rs.*	*Liabilities Rs.*
Stock	10,000		20,000	
Debtors	30,000		30,000	
Payment in Advance	2,000		—	
Cash in hand	20,000		15,000	
Sundry Creditors		25,000		30,000
Acceptances		15,000		12,000
Bank Overdraft		—		5,000
	62,000	40,000	65,000	47,000

Sales amounted to Rs.3,50,000 in the first year and Rs.3,00,000 in the second year.

You are required to comment on the solvency position of the concern with the help of

SOLUTION :

Short-term Solvency Analysis

(1) Current Ratio = $\frac{\text{Current Assets}}{\text{Current Liabilities}}$

Year I : $\frac{10{,}000 + 30{,}000 + 2{,}000 + 20{,}000}{25{,}000 + 15{,}000} = \frac{62{,}000}{40{,}000}$

= **1.55 : 1**

Year II : $\frac{20{,}000 + 30{,}000 + 15{,}000}{30{,}000 + 12{,}000 + 5{,}000} = \frac{65{,}000}{47{,}000}$

= **1.38 : 1**

(2) Liquid or Quick Ratio = $\frac{\text{Liquid Assets}}{\text{Liquid Liabilities}}$

Year I : $\frac{30{,}000 + 20{,}000 + 2{,}000}{25{,}000 + 15{,}000} = \frac{52{,}000}{40{,}000}$

= **1.30 : 1**

Year II : $\frac{30{,}000 + 15{,}000}{30{,}000 + 12{,}000 + 5{,}000} = \frac{45{,}000}{47{,}000}$

0.96 : 1

(3) Inventory Turnover Ratio = $\frac{\text{Net Sales}}{\text{Average Inventory}}$

Year I : $\frac{3,50,000}{10,000} = \mathbf{35:1}$

Year II : $\frac{3,00,000}{15,000} = \mathbf{20:1}$

(4) Inventory Current Assets Ratio = $\frac{\text{Inventory}}{\text{Total Current Assets}} \times 100$

Year I : $\frac{10,000}{62,000} \times 100 = \mathbf{16\%}$

Year II : $\frac{20,000}{65,000} \times 100 = \mathbf{31\%}$

(5) Average Collection Period = $\frac{\text{Trade Receivables}}{\text{Net Credit Sales}} \times$ No. of Working Days

Year I : $\frac{30,000}{3,50,000} \times 365 = \mathbf{31.3\ days}$

Year II : $\frac{30,000}{3,00,000} \times 365 = \mathbf{36.5\ days}$

Long-Term Solvency Analysis

(1) Debt Equity Ratio = $\frac{\text{External Equities}}{\text{Internal Equities}}$

Year I : $\frac{25,000 + 15,000}{62,000 - 40,000} = \frac{40,000}{22,000} = \mathbf{1.82:1}$

Year II : $\frac{30,000 + 12,000 + 5,000}{65,000 - 47,000} = \frac{47,000}{18,000} = \mathbf{2.61:1}$

(2) Proprietary Ratio is = $\frac{\text{Shareholder's Equities}}{\text{Total Equities}}$

Year I : $\frac{22,000}{62,000} = \mathbf{0.35:1}$

Year II : $\frac{18,000}{65,000} = \mathbf{0.28:1}$

Problem 13:

Following is the summarised Balance Sheet of a concern as at 31st December:

BALANCE SHEET as on 31st December

Liabilities	*Rs.*	*Assets*	*Rs.*
6% Preference Share Capital	1,50,000	Goodwill	20,000
Equity Share Capital	2,50,000	Land & Buildings	2,50,000
General Reserve	20,000	Machinery	1,75,000
Profit and Loss	15,000	Furniture	10,000
5% Debentures	1,00,000	Stock	90,000
Sundry Creditors	28,000	Sundry Debtors	21,000
Bills Payable	12,000	Cash at Bank	5,000
		Preliminary Expenses	4,000
	5,75,000		5,75,000

Other information :

Total sales Rs. 4,00,000 : 20% of which is made on credit. Gross Profit and Net Profit (after tax) for the year ended amounted to Rs. 80,000 and Rs. 20,000 respectively.

Comment on the Financial condition of the concern.

SOLUTION :

(1) *Current Ratio* :

$$\frac{\text{Current Assets}}{\text{Current Liabilities}} = \frac{\text{Rs. 1,16,000}}{\text{Rs. 40,000}} = \mathbf{2.9 : 1}$$

(2) *Liquid Ratio* :

$$\frac{\text{Liquid Assets}}{\text{Liquid Liabilities}} = \frac{\text{Rs. 26,000}}{\text{Rs. 40,000}} = \mathbf{0.65 : 1}$$

(3) *Proprietary Ratio* :

$$\frac{\text{Proprietors's Equity}}{\text{Total Assets}} = \frac{\text{Rs. 4,11,000}}{\text{Rs. 5,51,000}} = \mathbf{0.75 : 1}$$

(4) *Fixed Assets Proprietorship Ratio* :

$$\frac{\text{Fixed Assets}}{\text{Proprietors Equity}} = \frac{\text{Rs. 4,35,000}}{\text{Rs. 4,11,000}} = \mathbf{1.06 : 1}$$

(5) *Debt-Equity Ratio* :

$$\frac{\text{Total Debt}}{\text{Proprietors Equity}} = \frac{\text{Rs. 1,40,000}}{\text{Rs. 4,11,000}} = \mathbf{0.34 : 1}$$

(6) *Capital Gearing Ratio* :

$$\frac{\text{Equity Share Capital}}{\text{Pref. Share Capital + Debentures}} = \frac{\text{Rs. 2,50,000}}{\text{Rs. 2,50,000}} = \mathbf{1 : 1}$$

(7) *Gross Profit Ratio* :

$$\frac{\text{Gross Profit}}{\text{Sales}} \times 100 = \frac{\text{Rs. 80,000} \times 100}{\text{Rs. 4,00,000}} = \mathbf{20\%}$$

(8) *Net Profit Ratio* :

$$\frac{\text{Net Profit}}{\text{Sales}} \times 100 = \frac{\text{Rs. 20,000}}{\text{Rs. 4,00,000}} \times 100 = \mathbf{5\%}$$

(9) *Stock-Turnover Ratio*:

$$\frac{\text{Cost of Goods Sold}}{\text{Average Stock}} = \frac{\text{Rs. 3,20,000}}{\text{Rs. 90,000}} = \textbf{3.6 Times}$$

(10) *Debtors' Turnover Ratio* :

$$\frac{\text{Debtors}}{\text{Credit Sale}} \times 365 = \frac{\text{Rs. 21,000}}{\text{Rs. 80,000}} \times 365 = \textbf{96 Days}$$

(11) *Return on Proprietors' Fund* :

$$\frac{\text{Net Profit (after Tax)}}{\text{Proprietors – Fund}} = \frac{\text{Rs. 20,000}}{\text{Rs. 4,11,000}} = \textbf{0.05 : 1}$$

(12) *Turnover to Fixed Assets Ratio* :

$$\frac{\text{Turnover}}{\text{Fixed Assets}} = \frac{\text{Rs. 4,00,000}}{\text{Rs. 4,35,000}} = \textbf{0.92 : 1}$$

Workings :

1. *Current Assets*:

Stock	Rs. 90,000
Debtors	21,000
Cash at Bank	5,000

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